

YaP Combat (M2)

Custom PvE (and optional PvP) combat for the YaP MMO stack. Ships as `yap-combat.jar` with `CombatService` on Bukkit `ServicesManager`.

Custom HP model

Choice: ratio-scaled vanilla bar (not a separate boss bar).

Layer	Behaviour
Internal	Integer custom HP in <code>yap_combat_state.current_hp</code>
Max HP	<code>base-hp + hitpoints_level × hp-per-hitpoints-level</code>
Display	Vanilla hearts always max 20; current = $(\text{currentHp} / \text{maxHp}) \times 20$

The config key `custom-hp.hearts-display: 10` documents RS-style granularity (10 custom HP $\approx$ one logical “heart” in tuning docs). Display still uses the ratio model so high HP levels (99 HP skill) fit the vanilla hotbar.

Alternative not used: separate boss-bar HP overlay.

Damage pipeline

1. `EntityDamageByEntityEvent` — vanilla damage **cancelled**
2. Hit roll on entity thread via `YapSched.entity`
3. `maxHit = floor((strength + gear + pot) × level-factor)`
4. Hit if `attackRoll > defenceRoll` (RS-lite)
5. Damage `0...maxHit`, minimum `formula.min-damage-on-hit` on successful hit
6. Optional **critical hits** (`formula.crit-chance`, `formula.crit-multiplier`)
7. **Knockback physics** on successful hits (`physics.*`)
8. **Combo meter** scales consecutive hit damage (`combo.*`)
9. **Status effects** — DOT/HOT/debuffs/CC from `status-effects.yml` and on-hit procs
10. **Projectile physics** — custom velocity, pierce, drop-off, headshots (`projectiles.*`)

Hit rolls run on the **async scheduler** (`CombatHitResolver`); damage application stays on entity threads (Folia-safe).

Mob defence/strength derive from mob max health (`pve.mob-defence-health-divisor`, default 20). Each combat damage point removes `pve.health-per-combat-point` vanilla HP (default 2 $\approx$ one heart) so low-level “Hit 1” still kills chickens/zombies.

Player stats come from `SkillService` (attack, strength, defence, hitpoints) plus gear from `items.yml`.

Gear

- `plugins/YaPCombat/items.yml` — Material → bonuses
- Optional PDC `yapcombat:yap_gear_tier` on items for tier rows under `tiers:`

Food & potions

- **Food:** right-click items in `food.yml` → heal custom HP; cooldown `food.cooldown-ticks`
- **Potions:** three types in `config.yml` (`attack`, `strength`, `defence`) — material match, boost + duration + per-type cooldown; visible in `/combat stats`

Combat XP

When `yap-skills` is loaded, XP is granted via `SkillService.addXp`:

Skill	Source
Attack	damage dealt / kill share
Strength	damage dealt / kill share
Defence	damage taken
Hitpoints	share of all combat XP

Shares configurable under `xp.*` in `config.yml`.

PvP

- `pvp: false` by default in `config.yml`
- **YaPGames override:** active minigame matches allow PvP via `GameServices.allowPvp()` even when global PvP is off
- When enabled, respects YaPPlayerdata claim flag `pvp` (`RegionFlag.PVP`)
- **Attack cadence:** `combat.attack-cooldown-ticks` / `combat.ranged-cooldown-ticks` prevent click-spam
- **Skill cache:** async skill-level snapshot refreshed on join and `SkillLevelUpEvent` (`combat.skill-cache-ttl-ms`)

Status effects

Definitions in `plugins/YaPCombat/status-effects.yml` :

Effect	Kind	Notes
bleed / poison / burn	DOT	stacking damage over time
freeze / stun	CC	movement slow; stun blocks attacks
weakness / vulnerable	debuff	lowers stats or raises damage taken
enraged / regen	buff	strength boost or heal over time

On-hit procs in `config.yml` (`on-hit-procs`) and spell `applies-effect` in `spells.yml` . View active effects via `/combat stats` .

Combo meter

Chain hits within `combo.window-ms` for + `combo.bonus-per-stack` damage per stack (up to `combo.max`). Shown in action bar (`8x COMBO +21%`). Resets on miss when `combo.reset-on-miss: true` .

Projectiles

Managed arrows/bolts get boosted velocity, optional pierce (scales with ranged level), distance drop-off, and headshot multiplier. Routed through the same hit pipeline as melee.

Commands

Command	Permission
<code>/combat</code> or <code>/combat stats</code>	<code>yapcombat.use</code>
<code>/combat reload</code>	<code>yapcombat.admin</code>
<code>/yapcombat admin sethp <player> <hp></code>	<code>yapcombat.admin</code>

Death

- `death.keep-inventory` — RS-lite keep items flag
- Respawn restores custom HP when `death.restore-hp-on-respawn: true`

See also

- [MMO_PHASES.md](#) — M2 acceptance checklist
- [PERMISSIONS.md](#) — `yapcombat.*` nodes